Problem Set-II (PH-2017), Relativity

Department of Physics

1. Event A happens at point (x=5, y=3, z=0) and at time t given by ct=15; Event B occurs at (x=10, y=8,z=0) and ct=5, both in system S.
(a) What is the invariant interval between the two points A and B ?
(b) Is there an inertial frame in which they occur simultaneously ? If so, find its velocity (magnitude and direction) relative to the frame S.
(c) Is there any inertial frame in which they occur at the same point ? If so, find this velocity relative to the frame S.

2. (a) Write out the matrix that describes a Galilean transformation for an object moving with a velocity v parallel to X-axis.
(b) Write out the matrix that describes a Lorentz transformation for an object moving with a velocity v parallel to Y-axis.

3. Show that $dS^2 = c^2dt^2 - dx^2 - dy^2 - dz^2$ is invariant under Lorentz transformation but not invariant under Galilean transformation.

4. Find the change in the energy of an electron if its speed in increased from 1.2×10^8 m/s to 2.4×10^8 m/s.

5. Find the speed of 0.1 MeV electron according to non-relativistic and relativisitc classical mechanics.

6. A particle of rest mass m_0 and kinetic energy $6m_0c^2$ strikes and sticks to an identical particle at rest. What is rest mass of the resultant particle.